

Replacing Transmission with Generation: A Generative AI-based Transmission Framework for Haptic Feedback in the Metaverse

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Abstract—The introduction of haptic feedback into the metaverse can significantly enhance users’ immersive experience. However, performing high-quality multimodal content (including visual, auditory, and haptic) rendering on a user’s head-mounted displays (HMDs) is constrained by limited computational resources and battery capacity. Offloading the computational components to cloud servers can significantly alleviate these limitations, but the stringent 1 ms latency requirement of haptic feedback has become the major challenge. Inspired by the human brain’s ability to imagine the potential haptic feedback from visual content, we propose a Generative Artificial Intelligence (GAI)-based haptic feedback transmission framework that generates future potential haptic feedback based on visual inputs before the physical contact occurs to address this challenge. Moreover, we also design a Deep Reinforcement Learning (DRL)-based decision module to dynamically determine the optimal temporal lead and quality level to guarantee that the haptic feedback can be generated on time while maintaining a high quality. Comprehensive evaluation results demonstrate that the proposed approach can guarantee the 1 ms latency requirement of haptic feedback and achieve a high Structural Similarity Index Measure (SSIM) compared to the baseline approaches.

Index Terms—Generative artificial intelligence, deep reinforcement learning, haptic transmission, metaverse.

I. INTRODUCTION

THE incorporation of haptic feedback enables the metaverse to provide multimodal services, which include visual, auditory, and haptic content to users, thereby enhancing their immersive experience [1], [2]. By equipping users with head-mounted displays (HMDs) and haptic devices, such as haptic gloves and haptic suits, users experience a more realistic and engaging virtual environment where they can see, hear, and feel virtual objects. However, generating high-fidelity haptic feedback that is synchronized with visual and auditory content requires substantial computational resources, and the computational component is usually deployed at HMDs [3]. Therefore, rendering high-quality visual and auditory content

and associated haptic feedback on users’ HDMs requires powerful processors and batteries, leading to increased device weight, heat generation, and energy consumption, which significantly compromise user comfort and battery life [4], [5].

To address this issue, offloading computing tasks from users’ devices to cloud servers allows the devices to significantly reduce their weight and extend their battery life [6]. However, transmitting haptic feedback through networks usually requires ultra-low latency (typically 1 ms) to ensure a seamless and immersive user experience [7]. Unfortunately, rendering multimodal content at cloud servers introduces long-distance latencies, which makes it extremely challenging to meet the 1 ms latency requirement for haptic feedback [8]. Moreover, haptic feedback is vulnerable to network fluctuations and environmental interference, which means data loss or retransmissions will significantly affect the real-time performance of haptic feedback [9]. Therefore, how to guarantee the 1 ms latency requirement for haptic feedback in long-distance transmissions has become a major challenge.

Inspired by the human brain’s ability to imagine the potential haptic feedback from visual cues (for instance, people can imagine how a glass surface would feel by simply seeing it), we believe that if the future potential haptic feedback can be generated before the contact occurs, the long-distance latency issue could be addressed. To realize this, we propose a Generative Artificial Intelligence (GAI)-based haptic transmission framework that generates future potential haptic feedback by using the Latent Diffusion Model (LDM) [10] before the contact occurs. Moreover, instead of offloading the computational components to cloud servers, we offload them to edge servers, which are closer to users and can provide low-latency services.

Besides, the users’ Quality of Experience (QoE) is strongly impacted by several decisions in the haptic feedback generation process. First, the temporal lead must be carefully selected: if it is too early than the contact time, the generated feedback may be inaccurate; while if it is too close to the contact time, the LDM may not have enough time to generate high-quality feedback due to its iterative nature. Second, the generation quality of the haptic feedback itself is a key factor, as higher quality requires more computation and longer generation time. To jointly optimize these two decisions under dynamic environments (including network conditions and edge server loads), we design a Deep Reinforcement Learning (DRL)-based decision module that determines the optimal

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temporal lead and selects the appropriate generation quality for each user. To the best of our knowledge, this is the first work that leverages GAI models to generate future potential haptic feedback prior to contact in dynamic environments, thereby addressing the 1 ms latency challenge of haptic feedback.

The main contributions of our work are summarized as follows:

- To address the 1 ms end-to-end latency from edge servers to users' devices, we propose a novel GAI-based haptic transmission framework that generates future potential haptic feedback based on visual inputs before the physical contact occurs.
- We propose an LDM-based haptic feedback generation module that can generate high-quality haptic feedback based on the extracted video frames related to the expected contact. This allows the generated haptic feedback to be transmitted to users in advance, effectively mitigating the end-to-end latency of haptic feedback.
- We design a Deep Reinforcement Learning (DRL)-based decision module to dynamically determine the optimal temporal lead and quality for generating haptic feedback in dynamic networks. This ensures that high-quality haptic feedback can be delivered in time under varying network conditions and edge server loads.
- Comprehensive evaluation experiments have been conducted, and the evaluation results demonstrate that the proposed approach outperforms baseline approaches in terms of the average Structural Similarity Index Measure (SSIM) and the average latency satisfaction rate.

The remainder of this paper is organized as follows: Section II discusses related work. Section III shows the details of how the problem is modeled. In Section IV, we demonstrate the details of the proposed framework. Section V illustrates the simulation settings and results. The paper is concluded in Section VI.

II. RELATED WORK

To address the 1 ms latency challenge of haptic feedback, various approaches have been proposed in recent years. In general, there are two directions to tackle this challenge: 1) optimizing communication frameworks and compression algorithms to reduce the end-to-end latency of haptic feedback; 2) generating haptic feedback locally instead of directly transmitting it.

A. Communication Frameworks and Compression Algorithms for Low-Latency Haptic Transmission

For the first research direction, authors in [11] proposed a new mobile crowdsensing architecture by combining Cloud-Radio Access Networks (C-RAN) with Device-to-Device (D2D) to meet the latency requirements demanded by 5G networks. Authors in [12] multiplexed multiple access schemes for different traffic flows, enabling haptic feedback with a latency of 2 ms and a reliability of 99.999%. Kizilkaya et al. [13] proposed a task-oriented prediction and communication co-design framework for haptic communications, in which a

predictor is designed for predicting the trajectories of the human operator. Based on the predicted trajectories, the robotic arms can perform accurate actions even if some command packets are delayed or lost. By introducing the just noticeable difference (JND) as a task-dependent metric, the framework optimizes bandwidth allocation and data rate to meet stringent latency and reliability requirements for haptic feedback.

To reduce the end-to-end latency of haptic feedback, high compression ratio haptic encoders are proposed in [14]–[17], which can effectively reduce the encoding time and compress haptic feedback into smaller packets. However, high compression ratio haptic encoders may cause a large compression latency, which may not satisfy the 1 ms latency requirement of haptic feedback, e.g., the encoding latency of the proposed codec in [17] is 7 ms. To address this issue, Shaw et al. [18] proposed a parallel compression and transmission approach for haptic applications. Specifically, the proposed approach sends a fraction of the data uncompressed, while simultaneously compressing the remaining data. In this way, the latency that is caused by compression can be reduced, which can reduce the end-to-end latency for haptic feedback. He et al. [19] proposed a self-supervised spatio-temporal graph mask-passing attention network to predict perceptual importance of various interaction points in multi-point tactile interaction scenarios. Based on the predicted perceptual importance, the proposed approach can adaptively compress haptic feedback to reduce its transmission latency. A more recent in [20] proposed a cross-modal haptic compression scheme that leverages visual semantics to guide haptic codec parameter optimization.

B. Generating Instead of Transmitting Haptic Feedback

Regarding the second research direction, AI models, especially Long Short Term Memory (LSTM) [21]–[24], have been widely used to predict the haptic feedback. For instance, Long and Short term Time-series Network (LSTNet) is leveraged in [23] to predict the haptic feedback in telemedicine. After the future haptic feedback is generated, it can be sent in advance to mitigate the transmission latency of haptic feedback. Wang et al. [24] utilized LSTM to generate future haptic feedback based on the current haptic feedback. Then, the generated haptic feedback is sent out in advance to mitigate its transmission latency. However, this approach does not consider the transmission and generation time of haptic feedback, making it hard to satisfy the 1 ms latency requirement. Mondal et al. [25] proposed a two-stage framework to forecast haptic feedback samples. A supervised binary classifier is trained as the first stage to determine whether the haptic feedback needs to be predicted. Then, a Reinforcement Learning (RL) model is trained as the second stage to predict the future haptic feedback accurately.

Except for predicting haptic feedback based on historical haptic feedback, some researchers have attempted to leverage GAI models to generate haptic feedback based on other modalities, such as video frames and audio signals. For example, Generative Adversarial Networks (GANs) are utilized in [26] to construct a cross-modal connection between vision and haptic. Then the proposed approach can generate proper haptic

feedback from visual inputs. Authors in [27] leveraged a Diffusion Model model to synthesize haptic feedback, which has a higher accuracy compared to GANs. However, these approaches can only generate the related haptic feedback when the contact occurs, which means they cannot generate the 1 ms latency requirement of haptic feedback since there are transmission and generation latencies.

C. Summary

However, existing approaches still face challenges in meeting the 1 ms transmission latency requirement for haptic feedback. The first direction of research only focuses on optimizing the network architecture and protocols to reduce the transmission latency of haptic feedback, but it is still difficult to meet the 1 ms latency requirement due to the long-distance transmission. Although the second direction of research points out a promising way to address this challenge. Either the LSTM-based approaches, or the GAI-based approaches are still insufficient to meet the 1 ms latency requirement. Because either the LSTM-based approaches or the GAI-based approaches can only generate the future haptic feedback after the contact occurs. Considering the transmission latency and generation time of haptic feedback, it is still challenging to meet the 1 ms latency requirement. Therefore, it is essential to generate future potential haptic feedback before the contact occurs, so that the generated haptic feedback can be sent out in advance to mitigate its transmission latency.

III. SYSTEM MODEL

In this section, we present the details of the system model, which includes the network model, content model, and transmission model.

A. Network Model

We consider a network consisting of a cloud server $\mathcal{C}$, an edge server E , and a set of users $\mathcal{U} = \{u_1, u_2, \dots, u_n, \dots, u_N\}$. Users are connected to the edge server via a 5G access network, which includes multiple Base Stations (BSs). The cloud server is connected to the access network through the core network. We denote the set of BSs as $\mathcal{D} = \{d_1, d_2, \dots, d_g, \dots, d_G\}$. The cloud server is responsible for providing original multimodal services for all users. The Edge server is deployed in the access network to provide low-latency services for users. Moreover, the edge server is also responsible for generating and transmitting haptic feedback to users. Each user is equipped with an HMD and a pair of haptic gloves to receive multimodal services, and they are connected to edge servers via BSs. The overall network architecture is illustrated in Fig. 1.

We define the allocated bandwidth for user u_n as B_{u_n} , and the maximum bandwidth that a BS d_g can provide is $B_{d_g}^{\max}$. Then, we can have the following constraint:

$$\sum_{u_n \in \mathcal{U}_{d_g}} B_{u_n} \leq B_{d_g}^{\max}, \forall u_n \in \mathcal{U}_{d_g}, \forall d_g \in \mathcal{D}, \quad (1)$$

where $\mathcal{U}_{d_g}$ is the set of users connected to BS d_g .

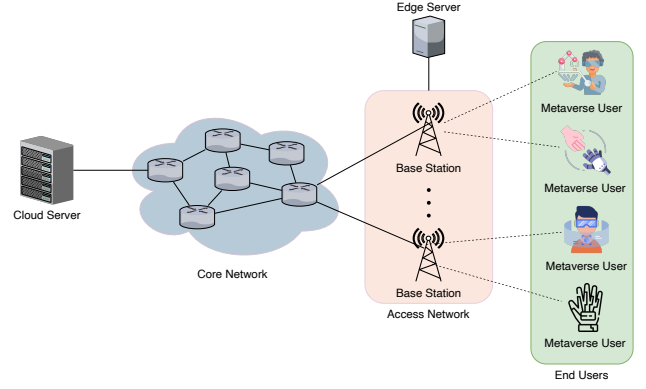


Fig. 1. The network model of the proposed GAI-based haptic feedback transmission framework, consisting of a cloud server, a core network, an access network with an edge server, and end users.

Compared to the cloud server, the edge server E has limited computational resources. Hence, we denote the total computational resources of the edge server E as $C_E^{\max}$, and the computational resources allocated to user u_n by edge server E as C_{u_n} . Then, we have the following constraint:

$$\sum_{u_n \in \mathcal{U}} C_n \leq C_E^{\max}, \forall u_n \in \mathcal{U}. \quad (2)$$

B. Content Model

The set of multimodal services that are provided by the metaverse can be defined as $\mathcal{S} = \{s_1, s_2, \dots, s_l, \dots, s_L\}$. Each service s_l can include multimodal content, such as visual, auditory, and haptic feedback content. Since visual and auditory content are delivered through conventional streaming pipelines, we focus on modeling the haptic feedback in this work. Therefore, we define that the haptic feedback of service s_l is represented as $\mathcal{H}(s_l)$ which can be further divided into multiple segments:

$$\mathcal{H}_{s_l} = \{h_{t_1, s_l}, h_{t_2, s_l}, \dots, h_{t_i, s_l}, \dots, h_{t_M, s_l}\} \quad (3)$$

where t_{end} is the end time of service s_l , and h_{t_i, s_l} is the haptic feedback segment at time t_i . Each haptic feedback segment has its own data size, quality, and latency requirement, which can be defined as follows:

$$h_{t_i, s_l} = \{d_{t_i, s_l}^h, q_{t_i, s_l}^h, l_{t_i, s_l}^h\} \quad (4)$$

where d_{t_i, s_l}^h is the data size of haptic feedback segment at time t_i , q_{t_i, s_l}^h is the quality of haptic feedback segment at time t_i , and l_{t_i, s_l}^h is the latency requirement of haptic feedback segment at time t_i .

C. Transmission Model

To guarantee the latency requirement of haptic feedback, we assume that the haptic feedback can only be provided by the edge server, as the long-distance transmission from the remote cloud server to users cannot satisfy the 1 ms latency requirement of haptic feedback. Thus, the end-to-end

latency of transmitting haptic feedback segment h_{t_i, s_l} from edge server E to user u_n can be expressed as follows:

$$L_{u_n}(h_{t_i, s_l}) = L_{u_n}^{trans}(h_{t_i, s_l}) + L_{u_n}^{proc}(h_{t_i, s_l}) + L_{u_n}^{prop}(h_{t_i, s_l}) + L_{u_n}^{queue}(h_{t_i, s_l}) \quad (5)$$

where $L_{u_n}^{trans}(h_{t_i, s_l})$ is the transmission latency, $L_{u_n}^{prop}(h_{t_i, s_l})$ is the propagation latency, $L_{u_n}^{proc}(h_{t_i, s_l})$ and $L_{u_n}^{queue}(h_{t_i, s_l})$ are the processing latency and queuing latency at the edge server. Notably, the processing latency $L_{u_n}^{proc}(h_{t_i, s_l})$ is determined by the computational resources C_{u_n} allocated to user u_n by the edge server.

To satisfy the latency requirement of haptic feedback, we have the following constraint:

$$L_{u_n}(h_{t_i, s_l}) \leq l_{t_i}(s_l), \quad \forall u_n \in \mathcal{U}, \forall s_l \in \mathcal{S} \quad (6)$$

where $l_{t_i}(s_l)$ is the latency requirement of haptic feedback segment h_{t_i, s_l} . Notably, the propagation latency and queuing latency are typically negligible in edge networks. In contrast, processing latency is the most significant component due to the substantial computational overhead (usually on the order of hundreds of milliseconds) required by GAI models for generating haptic feedback. Therefore, to mitigate the impact of processing latency, it is crucial to generate the future potential haptic feedback before the contact occurs.

IV. GAI-BASED HAPTIC FEEDBACK TRANSMISSION FRAMEWORK

In this section, we present the details of the proposed GAI-based haptic feedback transmission framework, which consists of a DRL-based decision module, a frame extraction module, and an LDM-based haptic feedback generation module. To further reduce the transmission latency of the generated haptic feedback, the proposed framework is deployed at the edge server. The overall architecture is illustrated in Fig. 2.

A. DRL-based Decision Module

To guarantee the user's QoE, two critical decisions need to be made in the haptic feedback generation process: 1) determining the optimal temporal lead t_p ; 2) selecting the optimal quality level q_l .

Obviously, the network conditions, such as available bandwidth, also have a strong impact on these two decision-making processes, as they can affect the transmission latency of haptic feedback. As mentioned before, the available computational resources of the edge server also have a significant impact on the processing latency of haptic feedback. Therefore, how to make these two decisions under dynamic network conditions and varying edge server loads can be formulated as a Markov Decision Process (MDP).

We use a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$ to represent the MDP, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $\mathcal{P}$ is the state transition probability, $\mathcal{R}$ is the reward function, and γ is the discount factor. Since the state transition probability $\mathcal{P}$ is unknown in real networks, we use the Trust Region Policy Optimization (TRPO) algorithm, which is a model-free DRL algorithm, to design the DRL-based decision module. After

training the DRL agent, the proposed DRL-based decision module can dynamically determine the optimal temporal lead t_p and select the optimal quality level q_l based on the current network conditions and the edge server loads.

1) *State Space*: The state $s_t \in \mathcal{S}$ observed by the DRL agent at time t includes the current available bandwidth bw_t and computational resources C_t of the edge server. To capture the short-term dynamics of the network and server load, we also include the previous time step's available bandwidth bw_{t-1} and computational resources C_{t-1} . Therefore, the state s_t can be represented as:

$$s_t = [bw_t, C_t, bw_{t-1}, C_{t-1}]. \quad (7)$$

2) *Action Space*: The action space $\mathcal{A}$ is a two-dimensional discrete space, where each action $a_t \in \mathcal{A}$ at time t consists of the temporal lead t_p and the generation quality level of haptic feedback q_l . Specifically, we define the generation quality level q_l as three levels: low, medium, and high. We denote the generation quality level selected at time t as $q_{l,t}$. Therefore, the action a_t can be represented as follows:

$$a_t = [t_p, q_{l,t}]. \quad (8)$$

Since the haptic feedback is represented as images in the *VisGel* dataset [26], the generation quality level q_l can be represented by the image resolution of the generated haptic feedback. Specifically, we define three quality levels: low (128×128), medium (192×192), and high (256×256). Additionally, we set the temporal lead t_p as 11 levels, ranging from 150 ms to 450 ms with a step size of 30 ms. Based on our experiments, the generation time of the haptic feedback is at least about 145 ms (see Fig. 4). Considering other types of latencies, such as transmission latency and processing latency, we set the minimal temporal lead as 150 ms. Therefore, the action space $\mathcal{A}$ contains 33 actions (11 temporal leads $\times 3$ quality levels).

Each quality level corresponds to a different number of diffusion steps Q in the LDM-based haptic feedback generation module. The higher the quality level, the more diffusion steps are required, resulting in a longer generation time. Therefore, the DRL agent needs a greater temporal lead. However, a greater temporal lead may lead to lower accuracy of the generated haptic feedback, which will degrade the user's QoE. Hence, there is a trade-off between the temporal lead and the generation quality level of haptic feedback.

3) *Reward Function*: To guarantee the user's QoE, the reward function should consider both the latency requirement and the quality of the generated haptic feedback. Hence, we define the reward at time t as follows:

$$r_t = \alpha \cdot \mathbb{I}(L_{E, u_n}(\hat{h}_{t_i, s_l}) \leq l_{t_i}(s_l)) + \beta \cdot q_{l,t} - \theta \cdot t_{p, \text{norm}}, \quad (9)$$

$$t_{p, \text{norm}} = \frac{t_p}{t_p^{\max}}, \quad (10)$$

where α, β and θ are the configurable weights. $\mathbb{I}(\cdot)$ is an indicator function that equals 1 if the latency requirement is satisfied, otherwise it equals 0. $L_{E, u_n}(\hat{h}_{t_i, s_l})$ is the end-to-end latency of transmitting the generated haptic feedback $\hat{h}_{t_i, s_l}$ from edge server E to user u_n , and l_{t_i, s_l} is the latency requirement of haptic feedback segment h_{t_i, s_l} . $q_{l,t}$ is the quality level

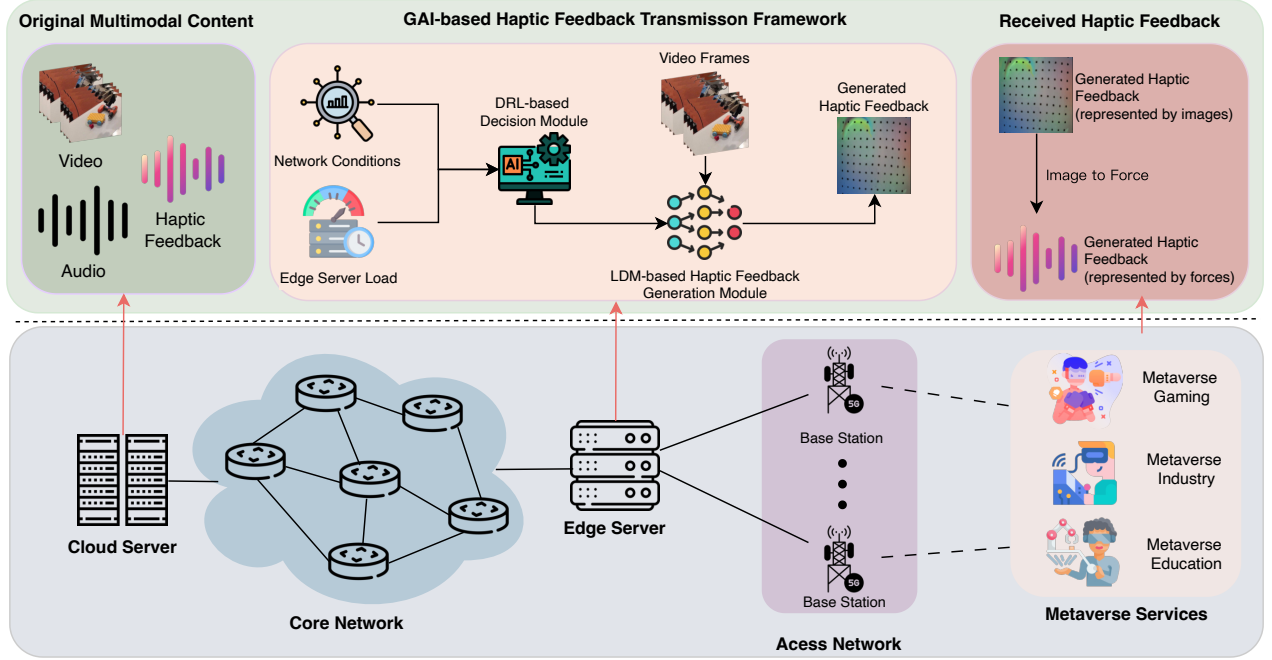


Fig. 2. The Overview of the Proposed GAI-based Haptic Feedback Transmission Framework.

of the generated haptic feedback at time t , represented by the SSIM value (ranging from 0 to 1). $t_{p,norm}$ is the normalized temporal lead, used to provide a negative reward (penalty) to prevent the agent from being overly conservative and selecting excessively large temporal leads.

B. Frame Extraction Module

The frame extraction module is responsible for extracting the appropriate video frames from the multimodal services. It needs to extract video frames before the actual contact occurs based on the temporal lead t_p determined by the DRL-based decision module. Specifically, if the contact occurs at time t_c , the frame extraction module will extract the video frame at time $t_c - t_p$. The extracted video frames are then sent to the LDM-based haptic feedback generation module to generate the future potential haptic feedback.

C. LDM-based Haptic Feedback Generation Module

LDM is one of the generative models that has recently attracted significant attention due to its ability to generate high-quality content. Compared to standard Diffusion Models (DMs), LDM performs the diffusion step by compressing the data into the latent space using a pre-trained autoencoder, which significantly reduces the computational time while maintaining generation quality.

In this work, we leverage the Latent Diffusion Model (LDM) to generate potential future haptic feedback based on the extracted video frames. During training, a video frame f_{t+t_p} is first encoded into a latent representation $z_0 = \mathcal{F}(f_{t+t_p})$ using a pretrained Variational AutoEncoder (VAE). The VAE maps the input frame into a latent space, where the

spatial resolution of the input frame determines the generation quality of the decoded output.

The latent diffusion process then iteratively refines the latent representation through Q denoising steps:

$$z_{q-1} = z_q - \epsilon_\theta(z_q, \mathcal{I}_t, q), \quad q = Q, Q-1, \dots, 1, \quad (11)$$

where z_q denotes the latent at step q , ϵ_θ is the learned denoising network, and $\mathcal{I}_t$ represents the conditional input, such as the generation quality level q_l determined by the DRL-based decision module.

After Q denoising steps, the final latent z_0 is decoded to generate the predicted haptic feedback:

$$\hat{h}_{t_i, s_i} = \mathcal{D}(z_0), \quad (12)$$

where $\mathcal{D}(\cdot)$ is the decoder that reconstructs the haptic feedback from the latent space. In our implementation, the generation quality is mainly controlled by the input resolution to the VAE (e.g., 128×128 , 192×192 , 256×256), and the number of diffusion steps Q . Increasing the resolution and diffusion steps can improve the quality of the generated haptic feedback, but it also increases computational cost and latency. Therefore, there exists a trade-off between generation quality and efficiency. More details about the LDM-based haptic feedback generation can be found in our previous work [28].

D. Training and Implementation

In the proposed framework, the LDM-based haptic feedback generation module and the DRL-based decision module are required to be trained before deployment. The following describes the training and implementation details of these two modules.

1) *Dataset*: We utilize the *VisGel* dataset [26] to train the LDM-based haptic feedback generation module. The dataset contains 12,000 touch actions, each comprising 300 pairs of temporally aligned video frames and corresponding haptic images. In each sequence, a robotic arm equipped with haptic sensors approaches and contacts various objects. Haptic images are only available after contact, while video frames prior to contact are used to predict future haptic feedback. The haptic images capture object deformation caused by the sensor, which can be further processed into contact area and normal force, enabling conversion into human-perceptible haptic feedback [29]. We use 80% of the data for training and 20% for testing.

2) *Hardware and Software Environments*: The training of both the LDM-based haptic feedback generation module and the DRL-based decision module is conducted on a server equipped with dual Intel Xeon Platinum 8352V CPUs and 4 RTX 4090 GPUs (each with 24 GB), using Python 3.12.2 and PyTorch 2.2.2 on Ubuntu 22.04.

3) *Hyperparameter Settings*: The key hyperparameters for training the LDM-based haptic feedback generation module are listed in Table I, and the key hyperparameters for training the DRL-based decision module are listed in Table II.

TABLE I
HYPERPARAMETER SETTINGS FOR LDM TRAINING

Parameter	Value
Batch Size B	32
Epochs E	100
Learning Rate η	2.0×10^{-6}
Initial Noise Variance (β_{start})	0.0015
Final Noise Variance (β_{end})	0.0195
Attention Resolutions R_{attn}	[4, 2, 1]
Transformer Depth D_{trans}	1
Codebook Size n_{embed}	16,384
Latent Dimension d_{embed}	4

TABLE II
HYPERPARAMETER SETTINGS FOR DRL TRAINING

Parameter	Value
Total Timesteps (T)	20,000,000
Learning Rate (η)	5×10^{-4}
Rollout Steps (N_{roll})	4,096
Discount Factor (γ)	0.995
Conjugate Gradient Steps (N_{CG})	25
CG Damping (λ_{CG})	0.1
Target KL Divergence (δ_{KL})	0.005

4) *Implementation Details*: To reduce the end-to-end latency, the proposed framework is deployed at the edge server, where the proposed framework can directly access the video frames and server load information. To further reduce the generation time of haptic feedback, we leverage DeepCache [30] that caches approximately unchanged features during the future sampling steps of the LDM to reduce the computational load. During inference, the LDM is configured with 40 denoising steps and a guidance scale of $s = 7.5$, to obtain the

trade-off between the generation quality and the computational efficiency.

5) *An Example of Generated Haptic Feedback*: Fig. 3 shows an example of the generated haptic feedback using the proposed framework with different temporal leads. When the temporal lead is less than 450 ms, the generated haptic feedback can accurately capture the contact area and force magnitude, which are represented by the shape and color of the haptic image, respectively. This indicates that the proposed framework can generate high-quality haptic feedback based on the video frames before the contact occurs.

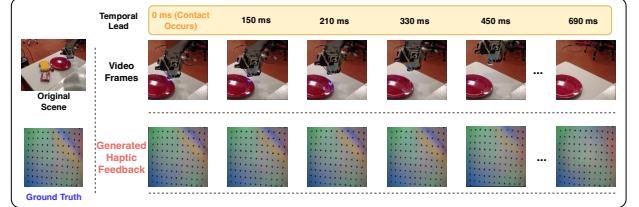


Fig. 3. An example of the generated haptic feedback from video frames under different temporal lead.

V. EVALUATION

In this section, we first present the evaluation metrics. Then, we introduce the simulation settings. Finally, we illustrate and analyze the evaluation results.

A. Simulation Settings

To simulate the dynamic network conditions, we generate random traffic to compete for the bandwidth of the access network. Moreover, we also add Gaussian noise to further simulate real-world wireless network fluctuations. Additionally, random computational tasks are generated to compete for the computational resources of the edge server.

B. Evaluation Metrics

Since haptic feedback is represented as images in this paper, we use **SSIM** [31] to evaluate the quality of the generated haptic feedback. The SSIM value ranges from -1 to 1, where 1 indicates perfect similarity.

We also use the **average latency satisfaction rate** (denoted by $\overline{S}_r$) to evaluate if the generated haptic feedback can meet the 1 ms latency requirement, which can be calculated as follows:

$$\overline{S}_r = \frac{1}{N} \sum_{l=1}^L \sum_{i=1}^M \mathbb{I}(L_{u_n}(h_{t_i, s_l}) \leq l_{t_i}(s_l)) \quad (13)$$

where N is the total number of generated haptic feedback segments, and $\mathbb{I}(\cdot)$ is an indicator function that equals 1 if the latency requirement is satisfied, otherwise it equals 0.

C. Baseline Approaches

We compare the proposed framework with the following baseline approaches:

- **Random Policy:** This approach randomly selects the temporal lead and generates the haptic feedback with a random quality level.
- **SSIM Greedy Policy:** In this approach, the edge server always selects the temporal lead and the generation quality level that obtains the highest SSIM, without considering the impacts of latency or computational load.
- **Success-Rate Greedy Policy:** Compared to the SSIM Greedy Policy, the edge server in this approach chooses the temporal lead and the generation quality level that maximizes the latency satisfaction rate.
- **Load Aware Policy:** This is a threshold-based approach that selects the temporal lead and quality level based on comparison between the threshold and the current edge server loads.
- **Bandwidth Aware Policy:** This is also a threshold-based approach that selects the temporal lead and quality level based on the comparison between the threshold and the current available bandwidth.

All baseline approaches leverage the same LDM-based haptic feedback generation module as our proposed framework to ensure a fair comparison. The main difference among these approaches is the decision-making process for selecting the temporal lead and quality level.

D. Evaluation Results and Analysis

We evaluate the performance of the proposed framework and baseline approaches under different network conditions and edge server loads. Specifically, we use the available bandwidth to represent the network conditions, and the GPU utilization rate to represent the edge server loads. To mitigate the impact of randomness, we conduct 5000 simulation runs for each comparison and calculate the average results.

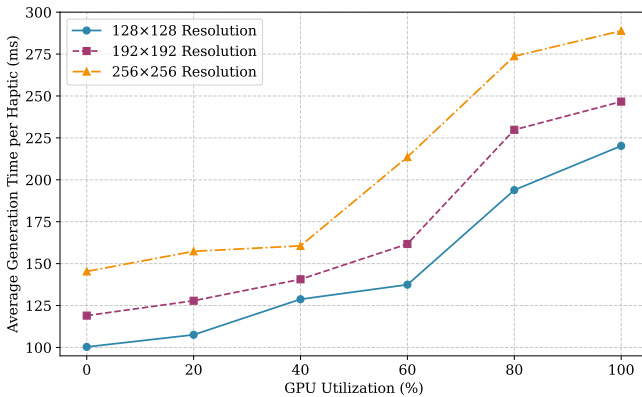


Fig. 4. Generation efficiency under varying GPU loads.

Fig. 4 illustrates the generation time of haptic feedback under different GPU utilization rates. As the GPU utilization rate increases, the generation time of all three quality levels also increases. Notably, when the GPU utilization rate is

higher than 40%, the generation time of all three quality levels increases significantly, e.g., the generation time of medium quality haptic feedback (192×192) increases from 160 ms to 275 ms when the GPU utilization rate increases from 40% to 80%. This indicates that the high GPU utilization rate can significantly degrade the generation efficiency of haptic feedback. Notably, to guarantee the 1 ms latency requirement of haptic feedback, the temporal lead should be greater than 150 ms at least, as the generation time of high-quality haptic feedback (256×256) is about 145 ms when the GPU utilization rate is 0%. Considering other types of latencies, such as transmission latency and processing latency, we set the minimal temporal lead as 150 ms in this work.

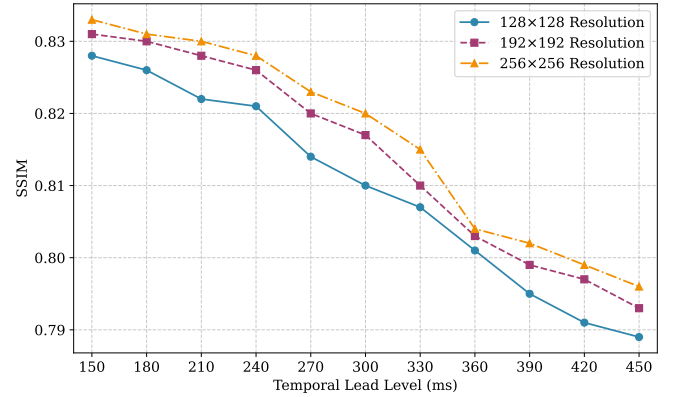
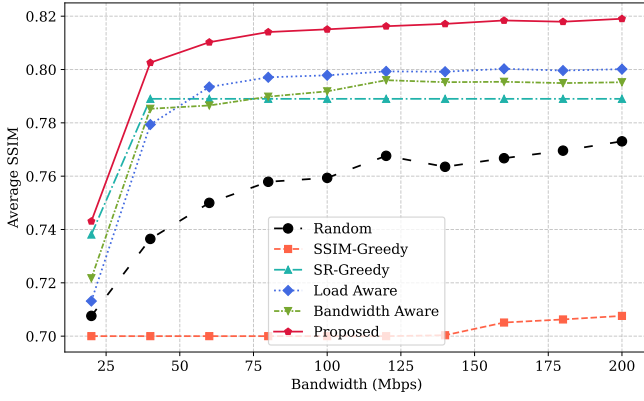


Fig. 5. Generation quality under different temporal lead.

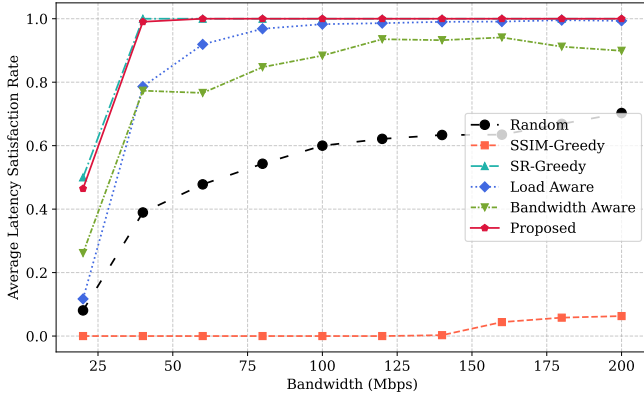
Fig. 5 demonstrates the generation quality of haptic feedback under different temporal leads. As the temporal lead increases, the generation quality of all three quality levels decreases. This is because a larger temporal lead means that the generated haptic feedback is based on video frames that are further away from the contact time, which can lead to lower accuracy of the generated haptic feedback. Notably, high generation quality of haptic feedback (e.g., 256×256) consistently achieves better SSIM values compared to lower generation quality haptic feedback (e.g., 128×128) across all temporal leads.

To evaluate the impacts of available bandwidth and edge server loads on the performance of all approaches, we conduct two sets of simulations by varying the available bandwidth from 20 Mbps to 200 Mbps and the edge server's GPU utilization rate from 0% to 100%, respectively. The results are presented in Fig. 6 and Fig. 7.

Fig. 6 shows the impact of available bandwidth on the performance of all approaches in terms of the average SSIM and the average latency satisfaction. From both Fig. 6 (a) and Fig. 6 (b), we can observe that the performance of all approaches improves as the available bandwidth increases. Our proposed framework consistently outperforms all baseline approaches in terms of both the average SSIM and the average latency satisfaction rate across all bandwidth levels. However, the impact of available bandwidth on the performance of all approaches becomes limited when the available bandwidth is higher than 40 Mbps. The reason behind this phenomenon is that the transmission latency of haptic feedback becomes



(a) Impact of available bandwidth on the average SSIM.

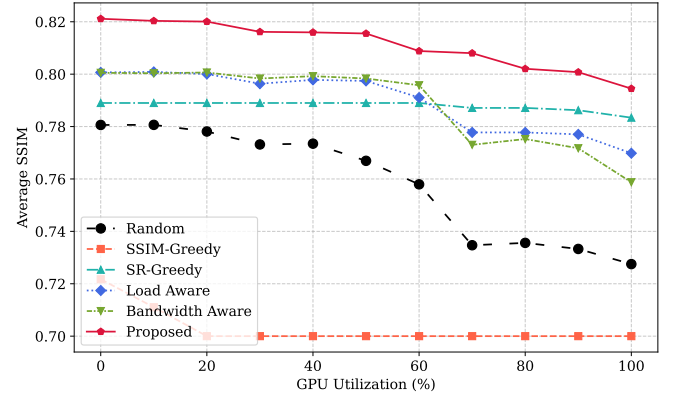


(b) Impact of available bandwidth on the average latency satisfaction rate.

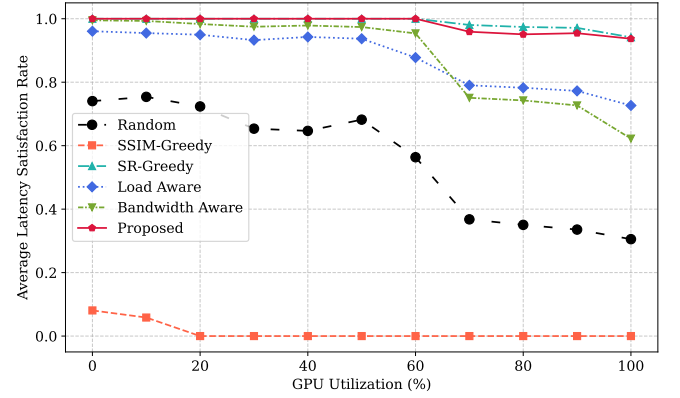
Fig. 6. Impact of available bandwidth on: (a). average SSIM; and (b). average latency satisfaction rate.

negligible compared to the processing latency when the available bandwidth is sufficiently high. Notably, the Load Aware policy can reach almost the same performance as our proposed framework when the available bandwidth is higher than 100 Mbps, but our proposed framework significantly outperforms it in terms of the average SSIM by 0.02 (which is considered perceptually significant) at least. Similarly, the Success Rate Greedy Policy can achieve the same performance as our proposed framework in terms of the average latency satisfaction rate, but its performance in terms of the average SSIM is much lower than that of our proposed framework. This demonstrates the effectiveness of our proposed framework in guaranteeing the latency satisfaction rate while improving the generation quality of haptic feedback. Another interesting observation is that the performance of the SSIM Greedy Policy is the worst in terms of both the average SSIM and the average latency satisfaction rate. This is because this approach always selects the highest quality level, which can lead to a high generation time and thus lead to a significantly high end-to-end latency of haptic feedback. As a result, the latency requirement of haptic feedback cannot be satisfied, and the received haptic feedback is often outdated, leading to a low average SSIM.

The impacts of edge server loads on the performance of all approaches in terms of the average SSIM and the average latency satisfaction rate are depicted in Fig. 7. From Fig.



(a) Impact of edge server loads on average SSIM.



(b) Impact of edge server loads on average latency satisfaction rate.

Fig. 7. Impact of edge server loads on: (a) average SSIM; and (b) average latency satisfaction rate.

7 (a), we can see that except for the Success Rate Greedy policy and the SSIM Greedy policy, all approaches experience a significant decrease in terms of the average SSIM as the edge server's GPU utilization rate increases to 70%. The possible reason is similar to the poor performance of the SSIM Greedy policy in Fig. 6 (a), i.e., a higher edge server loads lead to a longer generation time of haptic feedback, which can result in significantly outdated haptic feedback. This can be verified by Fig. 7 (b), where the average latency satisfaction rate of all approaches (except for the Success Rate Greedy policy and the SSIM Greedy policy) significantly decreases as the edge server's GPU utilization rate increases to 60%. Overall, our proposed framework consistently outperforms all baseline approaches in terms of both the average SSIM and the average latency satisfaction rate across all edge server load levels. Notably, our proposed framework can still maintain an average latency satisfaction rate of 95% even when the edge server's GPU utilization rate is as high as 100%, while the average SSIM is still higher than 0.79. This demonstrates the effectiveness of our proposed framework in guaranteeing the latency requirement of haptic feedback while improving the generation quality of haptic feedback under high edge server loads.

To evaluate the performance of all approaches over time, we conduct a 2000-second simulation and present the results in

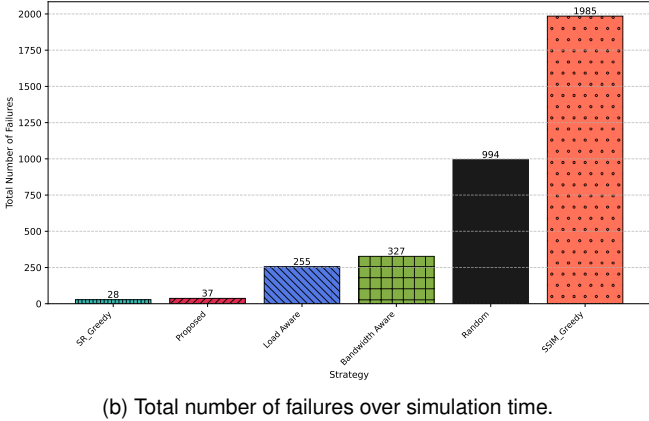
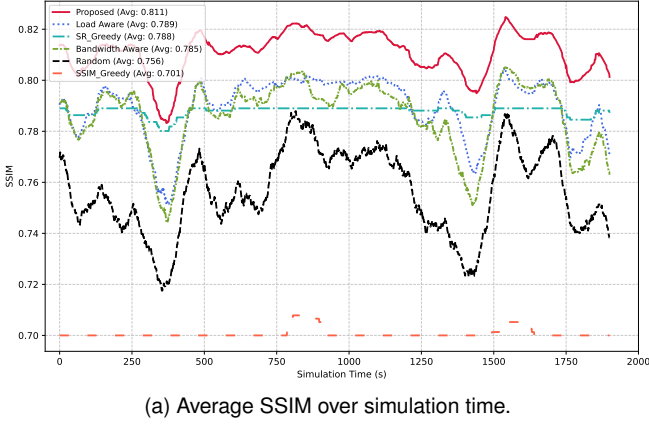


Fig. 8. Comparison of (a) average SSIM and (b) total number of failures over simulation time for all approaches.

Fig. 8. From Fig. 8 (a) and (b), we can see that our proposed framework consistently outperforms all baseline approaches in terms of both the average SSIM and the total number of failures over the entire simulation time. Obviously, the average SSIM is strongly affected by the satisfaction of the latency requirement, as the SSIM is calculated by comparing the received haptic feedback at time t with the ground truth haptic feedback at the same time t . If the received haptic feedback is outdated, i.e., it corresponds to an earlier contact time, e.g., $t - \Delta t$, not t . As a result, the SSIM score is significantly reduced since the feedback is mismatched with the current ground truth. Therefore, from Fig. 8 (b), we can observe that the proposed framework has the second-lowest total number of failures. Although the SR-Greedy policy has the lowest total number of failures, its average SSIM is the worst among all approaches, as shown in Fig. 8 (a). Therefore, the proposed framework can effectively balance the trade-off between satisfying the latency requirement and improving the generation quality of haptic feedback.

This contributes to the highest average SSIM among all approaches, as shown in Fig. 8 (a).

To evaluate the effectiveness of the proposed framework with different DRL models, we also compare the performance of five widely used DRL models: Deep Q-Network (DQN), Proximal Policy Optimization (PPO), Advantage Actor Critic (A2C), Asynchronous Advantage Actor-Critic (A3C), and

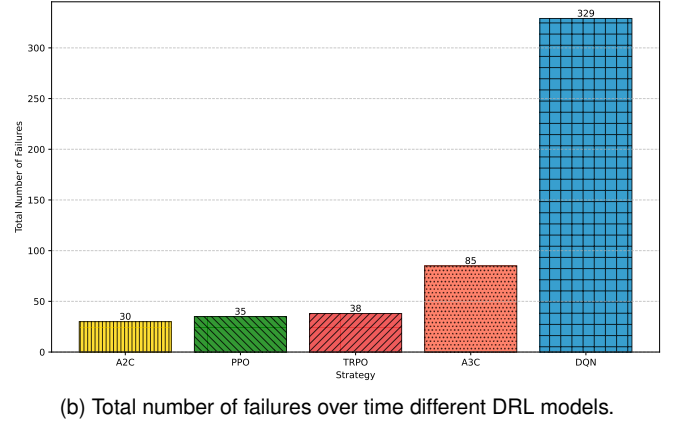
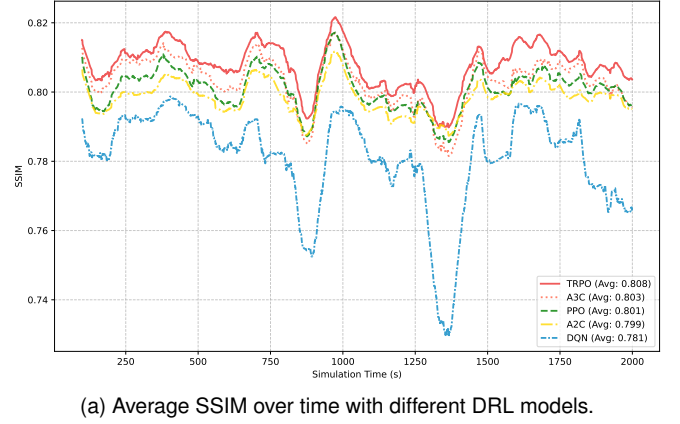


Fig. 9. Comparison of (a) average SSIM and (b) total number of failures over time with different DRL models.

TRPO over the long-term simulation time. Fig. 9 (a) depicts that TRPO consistently outperforms other DRL models in terms of the average SSIM. Notably, DQN has the worst performance, as it only works for simple scenarios. Fig. 9 (b) demonstrates that A3C, PPO, and TRPO have a close performance in terms of the total number of failures. Considering the overall performance in both the average SSIM and the total number of failures, we can conclude that TRPO is the most suitable DRL model for the proposed framework. The possible reason is that TRPO's trust-region constrained policy optimization helps to stabilize the training process and enhances robustness under dynamic bandwidth and edge server loads. This enables TRPO to achieve a better trade-off between exploration and exploitation compared to other DRL models.

VI. CONCLUSION AND FUTURE WORK

In this paper, we propose a novel GAI-based haptic feedback transmission framework to address the 1 ms latency challenge of haptic feedback in the metaverse. By generating future potential haptic feedback based on visual inputs before the actual contact occurs, the proposed framework can effectively mitigate the transmission latency of haptic feedback. Moreover, we designed a DRL-based decision module to dynamically determine the optimal temporal lead and generation quality level of generated haptic feedback under dynamic network conditions. We evaluated several DRL models, including

TRPO, PPO, A2C, and DQN, and found TRPO to be the most suitable for the proposed framework. Evaluation results demonstrated that the proposed approach can satisfy the 1 ms latency requirement for haptic feedback transmission while maintaining the quality of the generated haptic feedback under dynamic network conditions and edge server loads.

For future work, one potential direction is to improve the computational efficiency of GAI models to further reduce the generation time of haptic feedback. Moreover, the energy consumption could be considered in the design of the GAI models. Additionally, we also plan to leverage caching to cache basic haptic feedback to avoid redundant computation to further reduce the end-to-end latency of haptic feedback.

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